



भारतीय प्रौद्योगिकी संस्थान रुड़की Indian Institute of Technology, Roorkee
कंप्यूटर विज्ञान और अभियांत्रिकी विभाग Department of Computer Science and Engineering
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डॉ. पार्थ प्रतिम रॉय, सह प्राध्यापक

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CERTIFICATE OF INTERNSHIP

This is to certify that **Ms. Swarnim Sharma**, studying B.Tech. in Information Technology and Mathematical Innovation at Cluster Innovation Centre, University of Delhi, has successfully completed a three-month internship under my supervision at IIT Roorkee.

The internship, conducted from **1st December 2024 to 28th February 2025**, focused on the project titled "**EEG-Based Classification of Mental States During Meditation Using Machine Learning.**"

Swarnim designed and implemented a machine learning model to analyze EEG signals and classify attention states during meditation. The project involved preprocessing EEG data through artifact removal, extracting relevant features, and applying predictive modeling techniques. Additionally, Swarnim explored EEG connectivity, band power analysis, and statistical neuroscience methods to enhance classification accuracy and deepen the understanding of cognitive state recognition.

I wish her all the best in future endeavors.

Dr. Partha Pratim Roy
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